# SMART TRAFFIC MANAGEMENT SYSTEM FOR URBAN CONGESTION CONTROL

## Group No. 04

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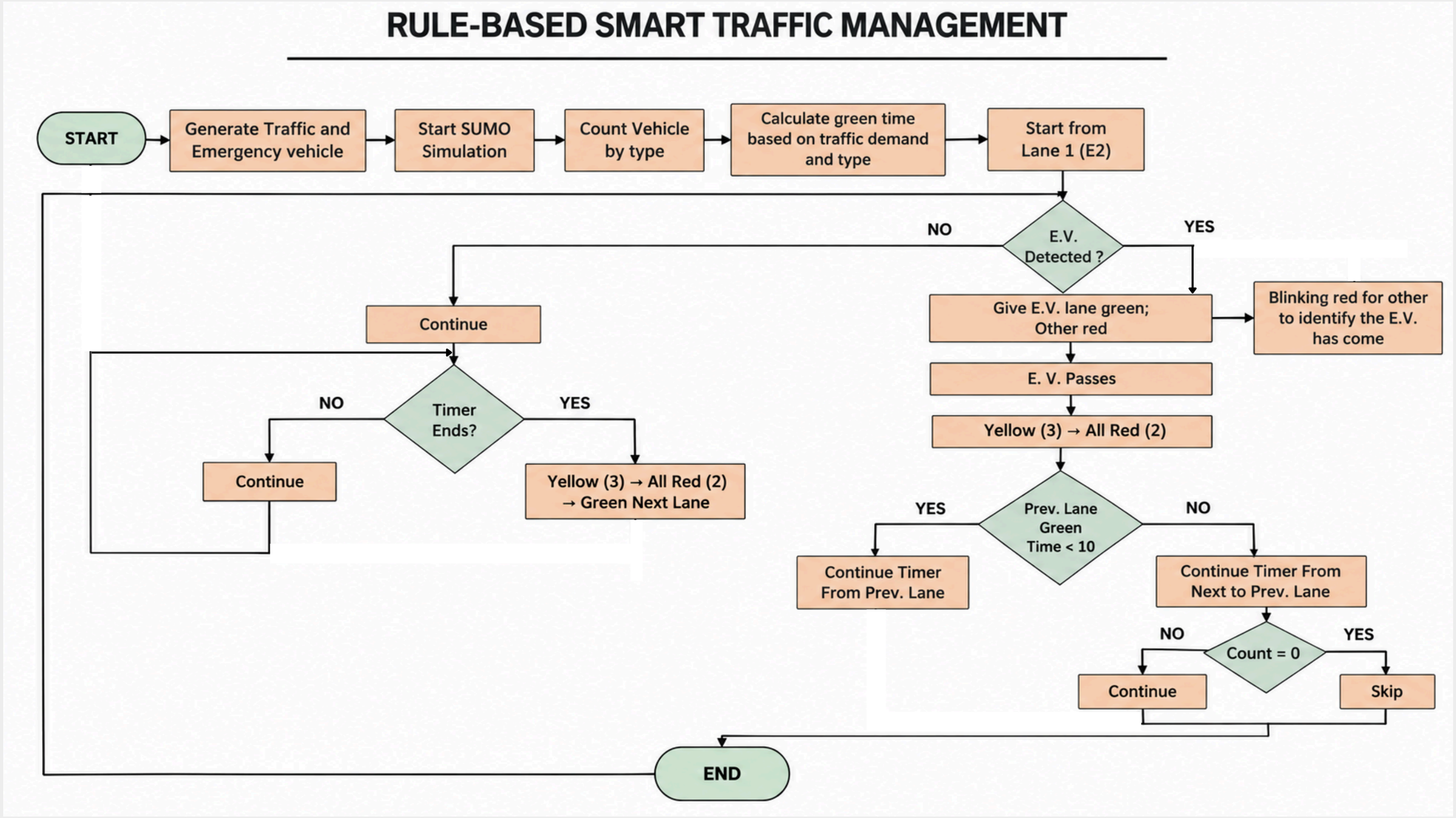
06 References




# PROGRESS VS PLAN

| Phase | Objective | 1-2 | 2-3 | 3-4 | 4-5 | 5-6 | 6-8 | 8-10 | 10-11 | 11-13 | 13-15 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Research and Planning | Problem analysis, research papers, system design | | | | | | | | | | |
| SUMO Setup and traffic simulation | Install SUMO, create roads, lanes and junction using NetEdit | | | | | | | | | | |
| Python + TraCI | Connect Python with SUMO and implement signal-control logic | | | | | | | | | | |
| Count + Emergency Priority | Timer based on congestion and Emergency Priority logic | | | | | | | | | | |
| More Features | More Features like - Type of vehicle, 2 Emergency vehicles at a time. | | | | | | | | | | |
| Final Testing & Documentation | Performance analysis, results and final report | | | | | | | | | | |

# TECHNICAL CONTENT DEPTH



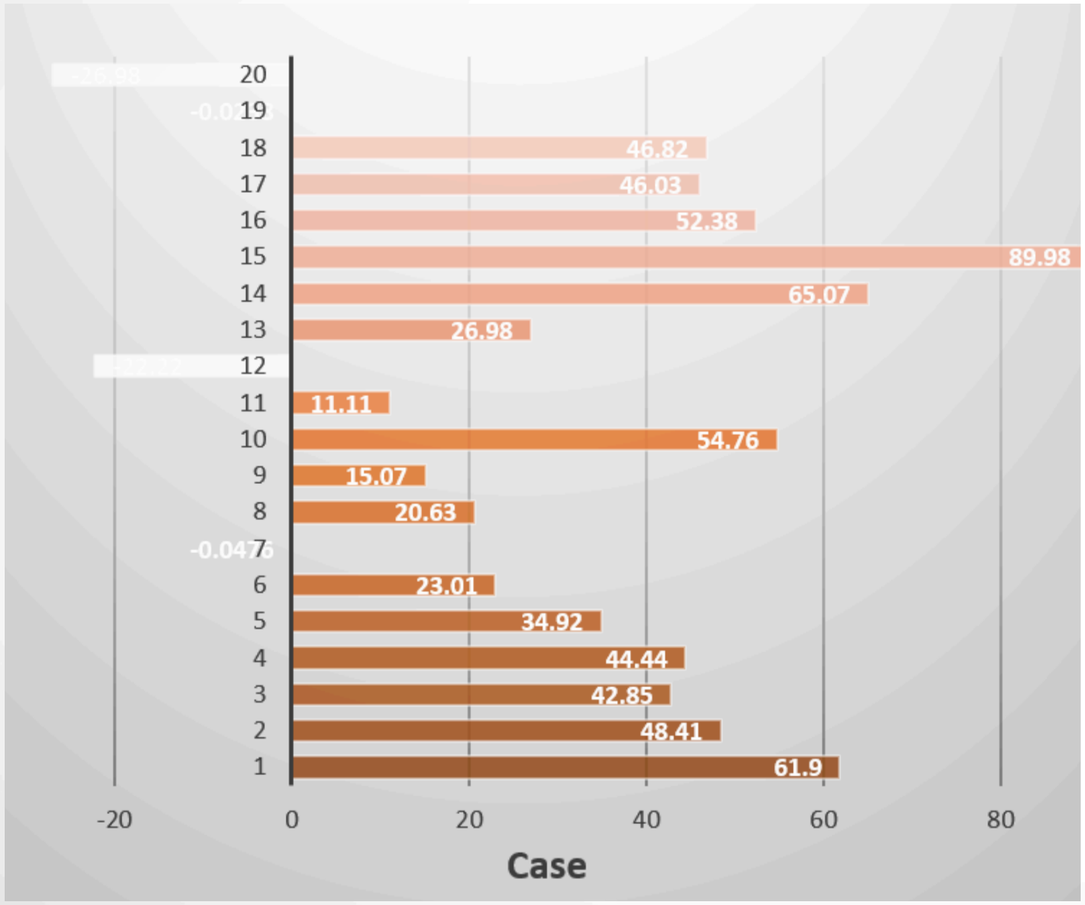
# FUNCTIONALITY ACHIEVED

- **Vehicle-Type-Based Dynamic Timing:** Calculates green signal duration based on lane-wise vehicle count and vehicle types, including cars, bikes, and heavy vehicles.
- **Emergency Vehicle Priority Control:** Detects emergency vehicles and prioritizes their lane by providing a green signal while keeping other lanes red.
- **Intelligent Signal Transition:** Applies yellow and all-red intervals after an emergency vehicle passes to manage safe transitions before resuming normal traffic control.
- **Adaptive Lane Switching:** Checks the previous lane's remaining green time and the next lane's vehicle count to decide whether to continue the timer or skip an empty



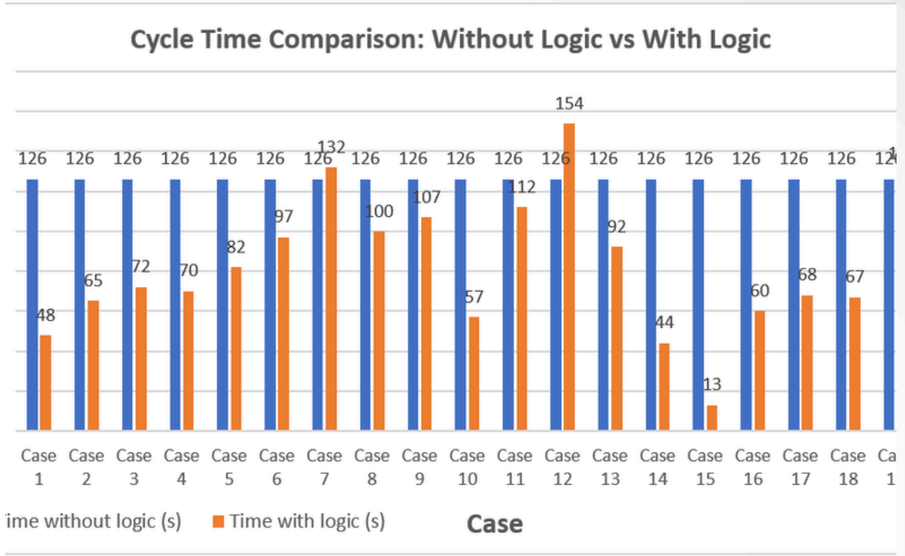



# Testing

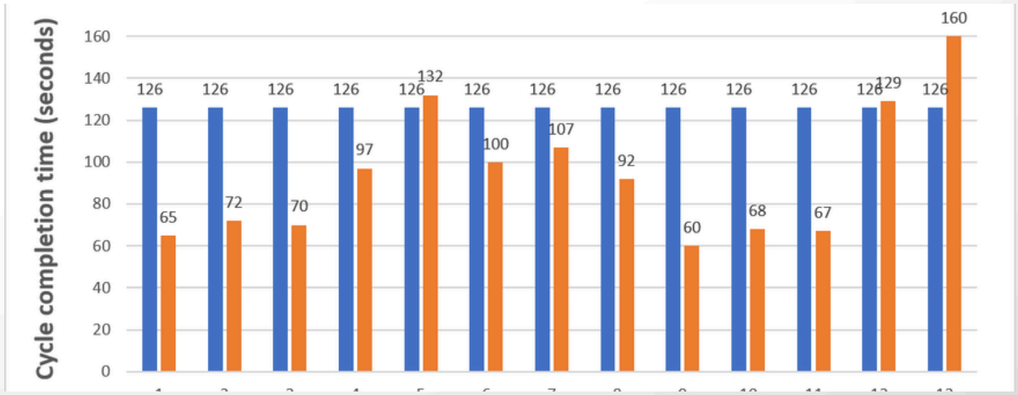



## Optimzation Percentage

## Cycle Time Comparison




## EV Handling: Cycle Time Comparison



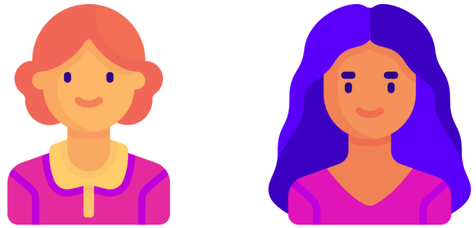
# INDIVIDUAL CONTRIBUTION



## Member 1

**Frontend & Environment :**

SUMO Netedit - road & lane creation , Traffic environment setup , Different signal formation (4 lane , 3 lane )



## Member 2 & 3

**Backend Logic :**

Python scripting , TraCI API integration , AI rule-based creation Traffic signal control logic



## Member 4

**Documentation :**

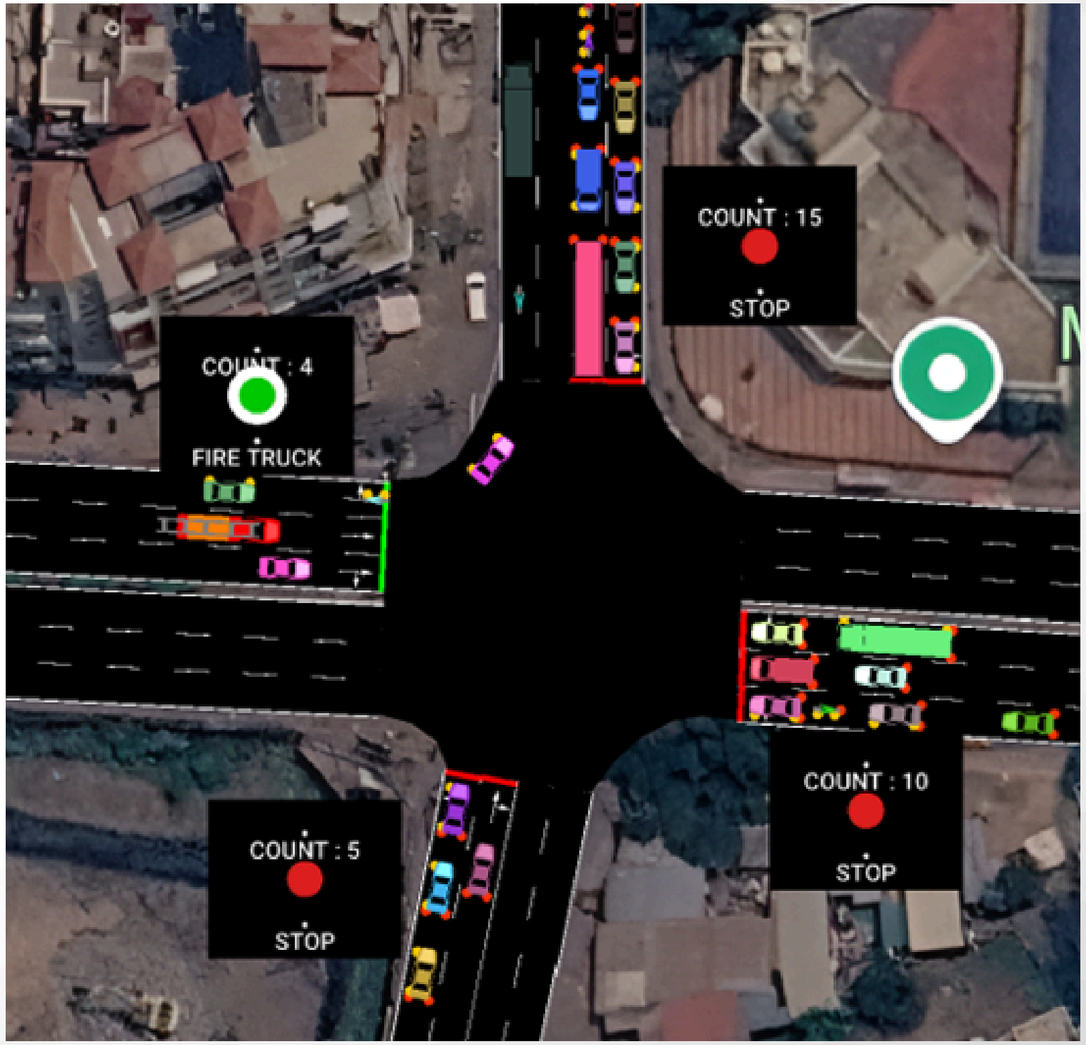
Team Co-ordination, Design documents Testing reports , Presentation creation, Research analysis

# Results & Analysis

The system adapts to changing traffic conditions to help reduce vehicle waiting time and congestion.

The system dynamically adjusts traffic signal timings based on vehicle count and traffic density.

Emergency vehicles are prioritized using predefined rules to support faster movement



# References


1. Adaptive Traffic Signaling Control Using SUMO Simulator (2025)
2. A SUMO-Based Simulation Framework for Intelligent Traffic Management System (2020)
3. AI-Powered Traffic Signal Control: A Deep Reinforcement Learning Framework for Urban Congestion Reduction (2025)
4. Enhancing Traffic Monitoring Systems through Image Analysis and Rule-Based Machine Learning (2025)
5. Development of the Rule-Based Approach to Traffic Management (2017)
6. Rule-Based Dynamic Traffic Management for Emergency Vehicle Routing: A Smart Infrastructure Approach (2019)
7. Dynamic Emergency Vehicle Prioritization in a Connected Urban Environment: A Rule-Based Approach (2024)
8. Traffic Intersection Simulation Using Turning Movement Count Data in SUMO: A Case Study of Toronto Intersections (2025)
9. AI-Based Traffic Control and Management Systems – A Synthesis of Macedonian Research Experiences (2025)
10. Computation and Optimization of Traffic Network Topologies Using Eclipse SUMO (2022)




# Thanks

**Here we are with our full Working Simulation on Smart Traffic Management !!!**

